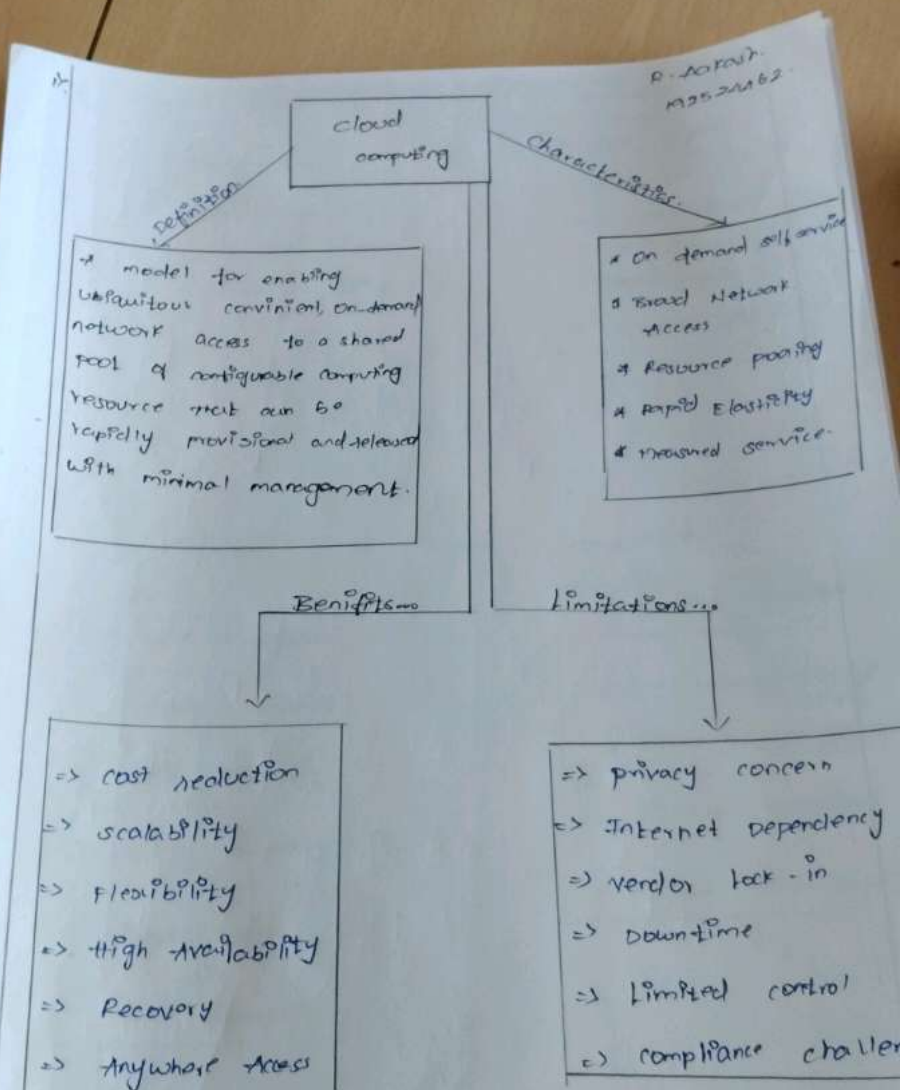




## Hardware Evolution

- ⇒ 1939 } vacuum tubes
- ⇒ 1941 }
- ⇒ 1946 ENIAC computer; 2<sup>nd</sup> gen
- ⇒ 1941 Intel 1004; 3<sup>rd</sup> Gen
- ⇒ 1971 MITS 8800 PC, 1<sup>st</sup> gen

FAAS.

## Software Evolution.

- ⇒ 1957, Sputnik.
- ⇒ SAGE
- ⇒ ARPANET (1967)
- ⇒ TCP/IP
- ⇒ Mosaic Browser
- ⇒ Mozilla (1994)

Evolution of cloud computing.

## Distributed system

- ⇒ cluster computing  
↓
- ⇒ Grid computing  
↓
- ⇒ utility computing  
↓
- ⇒ virtualization  
↓
- ⇒ cloud computing

## Service Delivery

- \* on demand self service
- \* Elastic resource provision
- \* measured service
- \* Global Access to ser

| IaaS                                           |              |
|------------------------------------------------|--------------|
| Delivers virtualized Infrastructure resources. |              |
| provider                                       | user         |
| Data center                                    | OS & backups |
| physical server                                | middle ware  |
| storage                                        | runtime      |
| Networking                                     | Application  |
| Virtualization                                 | data.        |

| PaaS                               |                |
|------------------------------------|----------------|
| Delivers platform for application. |                |
| provider                           | user           |
| Data centers                       | Application    |
| OS                                 | data           |
| Storage                            | security       |
| Database                           | Backups        |
| Networking                         | configuration. |

service Models...

| SaaS                 |               |
|----------------------|---------------|
| Delivers application |               |
| provider             | user          |
| Data center          | App Access    |
| storage              | manage user   |
| Runtime              | configuration |
| middle ware          | data input    |
| Networking           |               |

| XaaS               |                  |
|--------------------|------------------|
| Delivers anything. |                  |
| provider           | user             |
| Data center        | specific service |
| storage            | configure        |
| Networking         | manage.          |
| Virtualization     | security.        |

